

Understanding ASHE and HMRC reference numbers

Contents

Contents.....	1
1. ASHE files	2
2. HMRC lookup file	3
3. HMRC data – Frameworks	6
4. HMRC data - EoY file	8
5. HMRC data – Employments	9

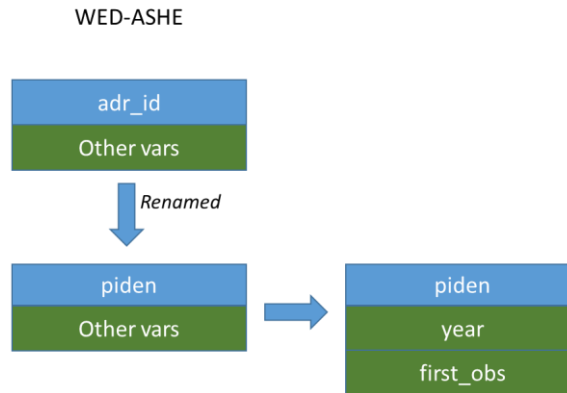
Note: Coding in here come from QA_01_confirm_pidens_vxx.do

1. ASHE files

ASHE files come as CSVs with an index:

Name	Format	Range	Notes
adr_id	yyyynnnnnnnnn 12 digits double	1999nnn – 2018nnnn (current dataset is up to 2018)	First four digits are year when that NINo first observed in ASHE
<i>plus other variables</i>			

The 'adr_id' is renamed as 'piden' in the ASHE code. Additionally, a small file "ashe_refs" is created with just year, piden and year of first observation, to allow testing agreement with the HMRC files:



Created files:	<i>ASHE files</i> <i>Created in core WED code</i>	<i>ashe_refs</i> <i>Used for checking matches</i>
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2. HMRC lookup file

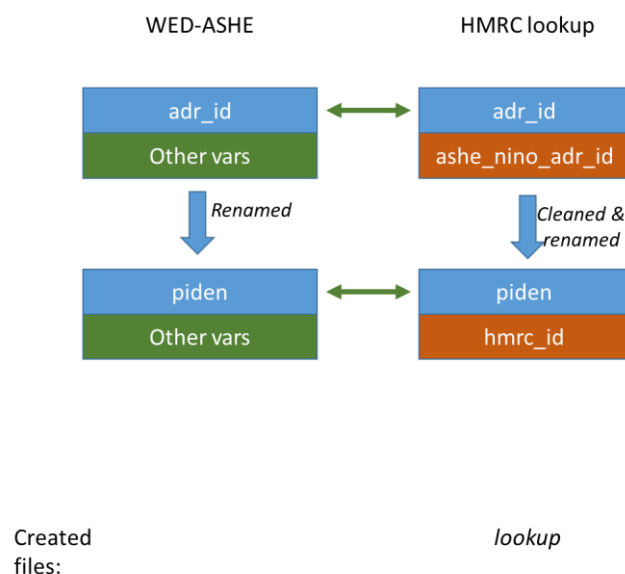
The 'lookup file' supplied by ONS contains

Name	Format	Range	Notes												
adr_id	[text_]nnnnnnnnnnnn 12 digits+text string	When text is removed ref numbers are this long: <table><tr><th>N. digits</th><th>N. cases</th></tr><tr><td>8</td><td>37</td></tr><tr><td>9</td><td>209</td></tr><tr><td>10</td><td>1,333</td></tr><tr><td>11</td><td>15,220</td></tr><tr><td>12</td><td>594,275</td></tr></table>	N. digits	N. cases	8	37	9	209	10	1,333	11	15,220	12	594,275	Some of the long numbers WED-ASHE-like but not consistently. Pre-digit text is one of “adr_ashe_w2_” “ashe_nes_” - Nothing
N. digits	N. cases														
8	37														
9	209														
10	1,333														
11	15,220														
12	594,275														
ashe_nino_adr_id	2021nnnnnnnn														

	11 digits double		
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So, adr_id looks (once text is removed) to be the same number in the ASHE file, generally. There are some duplicates in adr_id but these are all where the same 12-digit number has text in front or no text in front – there are no cases where the number part of adr_id is associated with a different ashe_nino_adr_id.

So it looks like we should be able to merge on adr_id (which we rename as piden), and linking it with the internal reference code ashe_nino_adr_id (which we rename as hmrc_id):



The match rates are not great. The match rate between all ASHE piden in the period 1997-2022 and the lookup file is around 68% (equivalent to 452,288 distinct piden). There are around 28% of the ASHE piden are missing from the HMRC look-up. However, this figure stays around 20% for the period between 2014-2019. It's okay that the HMRC IDs are many more, as they cover the full range of potential NINOs.

year of observation	Lookup only	ASHE only	Both	Total
2014	0	40,615	140,426	181,041
		22.43%	77.57%	100%
2015	0	38,620	140,360	178,980
		21.58%	78.42%	100%
2016	0	35,908	138,996	174,904
		20.53%	79.47%	100%
2017	0	34,471	140,310	174,781
		19.72%	80.28%	100%
2018	0	33,240	142,964	176,204
		18.86%	81.14%	100%
2019	0	31,388	141,479	172,867
		18.16%	81.84%	100%
Total	158,786	214,242	844,535	1,217,563
	Acceptable Should be more HMRC refs	Not acceptable ASHE refs should be a subset of HMRC refs	Acceptable Should be in both	

However, this is because some ASHE NINOs are much older than HMRC data. If we limit the link to employees that first appear in ASHE in 2015 and 2019 (and if we limit that to first observation only:

Year	ASHE only	Both	Total
2015	433	12,844	13,277
	3.26%	96.74%	100.00
2016	539	18,990	19,529
	2.76%	97.24%	100.00
2017	601	24,237	24,838
	2.42%	97.58%	100.00
2018	761	29,356	30,117
	2.53%	97.47%	100.00
2019	720	29,750	30,470
	2.36%	97.64%	100.00
Total	3,054	115,177	118,231
	2.58%	97.42%	100.00

So better, although still of concern: all NINos should be in the lookup table, irrespective of when the owner first appeared in ASHE.

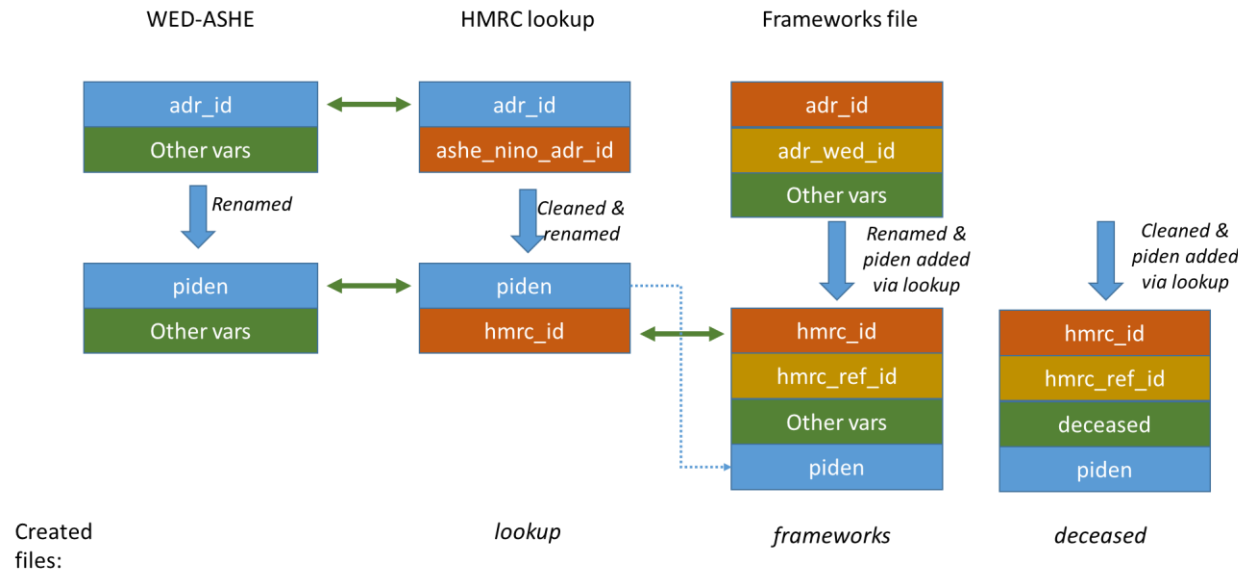
3. HMRC data – Frameworks

We now examine the EOY data as an example of HMRC linkage files. This contains:

Name	Format	Range	Notes
adr_id	2021nnnnnnnn 11 digits double		
adr_wed_id	Nnnnnnnnnn Up to 10 digits double	?	About half have 8 digits or less; they appear to be just reflecting dataset ordered (ie 1, 2, 3...) The other half all begin 859 and then have 7 digits
<i>plus other variables</i>			

adr_wed_id appears to represent a second internal HMRC reference number; it is uniquely associated with adr_id. At the moment we have retained both of these and renamed adr_wed_id as hmrc_ref_id but in the future we may be able to drop this variable.

As adr_id in this file matches the adr_ashe_nino_id in the lookup file we can add piden. We also use the clean-up process to extract the ‘deceased’ variable (the only item of interest in this file, it seems) in a more usable form, and save in a separate file linkable by piden:



Match rates are as follows:

HMRC references only in the frameworks file	1,037,970	21.23%	How is this the case?
HMRC references only in the lookup file	19,618	0.40%	Acceptable?
HMRC references in both	3,831,746	78.37%	Ideal

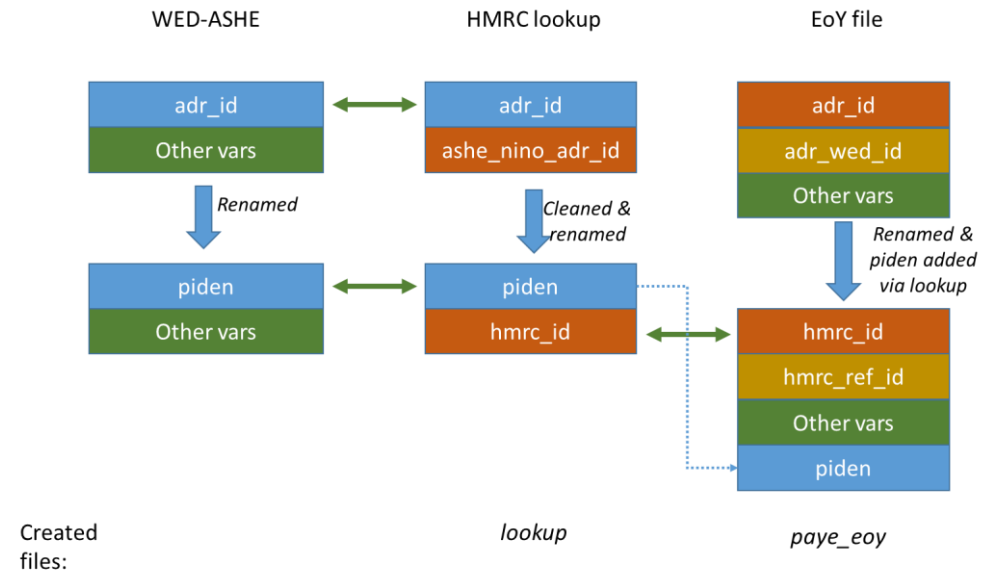
However, an individual can have multiple records. If we limit this to the numbers for each individual we get

HMRC references only in the frameworks file	349,484	36.38%	How is this the case?
HMRC references only in the lookup file	19,618	2.04%	Acceptable?

HMRC references in both	591,456	61.57%	Ideal
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4. HMRC data - EoY file

The same format as before with adr_id and adr_wed_id, so we rename.



Match rates this time show fewer non-matches between the data and the lookup file:

HMRC references only in the EoY file	149,129	4.4%	How is this the case?
HMRC references only in the lookup file	150,256	4.44%	Acceptable
HMRC references in both	3,087,278	91.16%	Ideal

This is for multiple entries as an individual can have several years of tax records. If we limit this to separate individuals we get

HMRC references only in the EoY file	32,893	5.11%	How is this the case?
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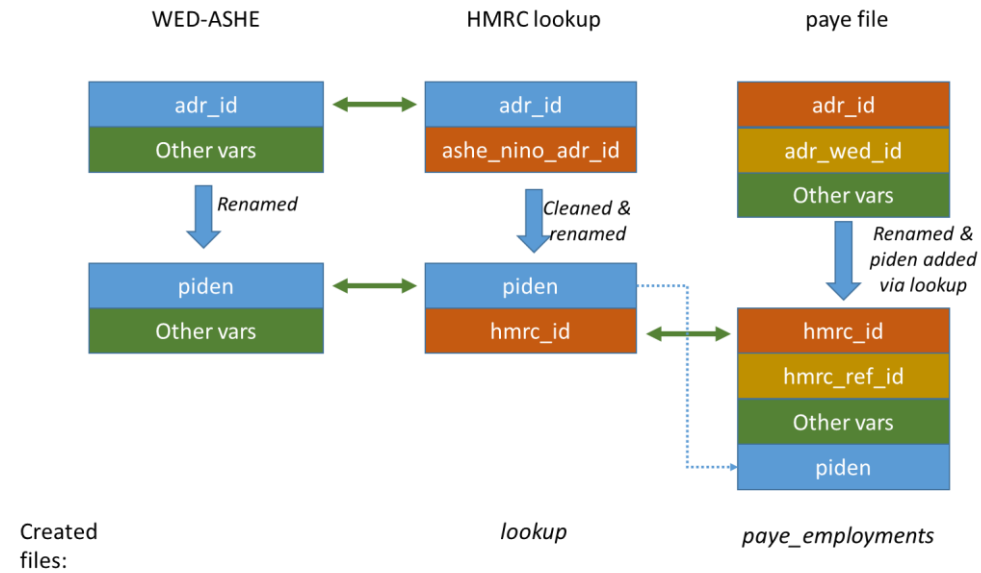
HMRC references only in the lookup file	150,256	23.33%	Acceptable
HMRC references in both	460,818	71.56%	Ideal

Looking at the coverage of the ASHE data, and restricting the data to one person (piden)/year in both the ASHE and HMRC data sets we get:

	ASHE and tax year			ASHE and tax year -1		
ASHE year	HMRC only	ASHE only	Both	HMRC only	ASHE only	Both
2015	414,842	35,361	206,841	410,380	36,735	211,303
2016	408,007	33,150	233,453	430,249	35,026	211,211
2017	443,187	31,570	205,152	410,673	32,935	237,666
2018	447,491	30,425	212,541	450,086	31,406	209,946
2019	425,843	28,830	239,050	450,895	30,402	213,998

5. HMRC data – Employments

As for EOY, this file has the two internal HMRC references so we link as before:



Match rates this time are similar for the EoY file. Looking at Employment/lookup overlap, and limiting to separate individuals we get

HMRC references only in the Employment file	39,411	6.06%	How is this the case?
HMRC references only in the lookup file	129,347	19.88%	Acceptable
HMRC references in both	481,727	74.06%	Ideal